

EXPERIMENT 20: TOWERS OF HANOI

AIM

To write a Prolog program to implement the Towers of Hanoi problem using recursion.

OBJECTIVE

- To understand recursion in Prolog.
- To solve the Towers of Hanoi problem.
- To move N disks from a source peg to a destination peg using an auxiliary peg.
- To follow the rule that a larger disk cannot be placed on a smaller disk.

ALGORITHM

1. Start.
2. If there is only one disk, move it directly from source to destination.
3. For N disks:
 - Move N-1 disks from source to auxiliary.
 - Move the largest disk from source to destination.
 - Move N-1 disks from auxiliary to destination.
4. Repeat recursively.
5. Display all moves.
6. Stop.

SOURCE CODE

% Towers of Hanoi

hanoi(1, Source, Destination, _) :-

write('Move disk from '),

write(Source),

write(' to '),

write(Destination),

nl.

hanoi(N, Source, Destination, Auxiliary) :-

N > 1,

N1 is N - 1,

hanoi(N1, Source, Auxiliary, Destination),

write('Move disk from '),

write(Source),

write(' to '),

write(Destination),

nl,

hanoi(N1, Auxiliary, Destination, Source).

Query

?- hanoi(3, left, right, middle).

OUTPUT

```
?- [hanoi].  
% hanoi.pl compiled 0.00 sec, 8 clauses
```

```
?- hanoi(3, left, right, middle).  
Move disk from left to right  
Move disk from left to middle  
Move disk from right to middle  
Move disk from left to right  
Move disk from middle to left  
Move disk from middle to right  
Move disk from left to right  
true.
```

```
?- 
```

RESULT

Move disk from left to right

Move disk from left to middle

Move disk from right to middle

Move disk from left to right

Move disk from middle to left

Move disk from middle to right

Move disk from left to right

true.

The Prolog program successfully solves the Towers of Hanoi problem using recursion. For 3 disks, it performs 7 moves, which is the minimum required ($2^3 - 1$).